ARAB GLOBAL CRYPTO EXCHANGE

Multi-Jurisdiction Platform Development

Requirements Specification & Architecture Brief

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1. Executive Summary & Objective

Arab Global Virtual Assets Services LLC SPC (AGV) is developing a full-featured, multi-jurisdiction crypto exchange ecosystem — the Arab Global Crypto Exchange (AGCE). The platform will serve retail and institutional clients across four distinct regulatory perimeters, each operating as a legally and technically isolated instance.

The four exchange instances are:

- AGCE India — FIU-IND registered, INR fiat-integrated, compliant with Indian VDA tax and AML frameworks
- AGCE Abu Dhabi — CMA regulated
- AGCE Dubai — VARA licensed, compliant with Dubai Virtual Assets Regulatory Authority framework
- AGCE Global — Serving loosely regulated jurisdictions with best-practice compliance baseline

All four instances share a common core technology platform (white-label engine) but are deployed in jurisdiction-isolated infrastructure with hard technical guardrails preventing cross-jurisdictional user comingling.

2. Multi-Jurisdiction Architecture

2.1 Architecture Philosophy

The platform follows a Shared Core / Isolated Instance model. A single versioned codebase is maintained centrally. Each jurisdiction deploys its own instance with jurisdiction-specific configuration modules applied at the infrastructure, application, and regulatory layer.

Shared across all instances	Matching engine, trading UI, order types, wallet infrastructure (Fireblocks), custody framework, hedging engine, admin panel framework, API gateway layer
Jurisdiction-specific	KYC/AML ruleset, fiat rails, supported asset list, product permissions (e.g. derivatives), fee schedules, regulatory reporting outputs, data residency configuration
Technically isolated	Database schemas, user identity namespace, IP geofencing, DNS/domain, TLS certificates, session tokens

2.2 Jurisdiction Isolation Guardrails

The following technical mechanisms must be implemented to ensure zero user comingling across exchange instances:

Identity & Account Namespace Isolation

- Each instance maintains an independent user identity store — no shared user database or SSO across instances
- Email/phone registration on one instance does not grant access to another — registration must be completed separately on each
- KYC verification is instance-specific; a verified user on AGCE India must re-verify on AGCE Dubai if they wish to access that platform

IP Geofencing & Access Control

- Each instance enforces IP-based country allow/block lists at the CDN and application layer
- AGCE India: allow IN IPs only, block all others by default
- AGCE Abu Dhabi: allow UAE + approved passport-holder IPs per CMA rules
- AGCE Dubai: allow UAE + approved passport-holder IPs per VARA rules
- AGCE Global: allow list of loosely regulated jurisdictions, block FATF high-risk countries, block India, UAE (prevent regulatory arbitrage)
- VPN/proxy detection layer (e.g. IPQualityScore or similar) applied at registration and login

Domain & Branding Isolation

- Each instance operates on a distinct domain or subdomain: in.arabglobal.exchange, abudhabi.arabglobal.exchange, dubai.arabglobal.exchange, global.arabglobal.exchange
- SSL certificates and HSTS pinning are instance-specific
- Cross-domain cookies and session sharing are explicitly disabled

Data Residency & Sovereignty

- AGCE India: all user PII and transaction data stored on AWS Mumbai (ap-south-1) — no cross-border data transfer
- AGCE Abu Dhabi: all data stored on UAE-region cloud infrastructure (Huawei Cloud UAE or AWS UAE)
- AGCE Dubai: data stored in UAE region with VARA-compliant data handling policies
- AGCE Global: data stored in a neutral jurisdiction (e.g. Singapore or Frankfurt) with GDPR-aligned privacy baseline
- Cross-instance data sharing is prohibited by architecture — no API calls from one instance to another's user data stores

Wallet & Fund Isolation

- Each instance operates under a separate Fireblocks workspace — funds from AGCE India users cannot be accessed via AGCE Dubai admin controls
- Deposit addresses are instance-specific — a BTC address issued on AGCE India is not valid on AGCE Global
- On-chain transactions are tagged with instance identifiers in internal ledgers for reconciliation
- INR fiat wallets exist only on AGCE India instance — no fiat wallet on other instances at launch

Regulatory Reporting Isolation

- Each instance generates jurisdiction-specific regulatory reports independently — FIU-IND for India, ADGM/CMA format for Abu Dhabi, VARA format for Dubai
- Compliance dashboard access is role-gated and instance-scoped — a compliance officer for AGCE India cannot view AGCE Dubai user data

2.3 Jurisdiction-Specific Compliance Matrix

Requirement	AGCE India	AGCE Abu Dhabi	AGCE Dubai	AGCE Global
Regulatory body	FIU-IND / SEBI (VDA)	CMA (ADGM)	VARA	Best-practice baseline
KYC provider	Didit + VKYC	Didit + Emirates ID	Didit + Emirates ID	Didit + passport

Requirement	AGCE India	AGCE Abu Dhabi	AGCE Dubai	AGCE Global
Fiat currency	INR (UPI / bank)	AED (bank transfer)	AED (bank transfer)	USDT/USDC only
Derivatives permitted	No (pending regulation)	Yes (CMA licensed)	Yes (VARA licensed)	Yes
AML reporting	FIU-IND STR/CTR	ADGM AML framework	VARA AML framework	FATF-based
Tax reporting	TDS 1% + ITR support	No crypto tax	No crypto tax	User-export only
Data residency	AWS Mumbai	UAE region	UAE region	Singapore / Frankfurt
Travel Rule	CryptoSwift	CryptoSwift	CryptoSwift	CryptoSwift
P2P trading	Yes (INR-USDT)	Restricted	Restricted	Yes

3. Core System Requirements

3.1 Matching Engine

- High-performance, low-latency CLOB matching engine (price-time priority)
- Real-time internal order matching — user vs user with no external dependency
- WebSocket-based live order book and trade updates
- Support for: Market, Limit, Stop-Limit, IOC (Immediate-Or-Cancel), FOK (Fill-Or-Kill), Post-Only orders
- Order book depth snapshots with configurable granularity
- Trade history, ticker, and OHLCV aggregation endpoints
- Engine must be implemented in Go or Rust for latency requirements
- Must NOT be a Binance or Bybit white-label wrapper — internal engine is mandatory

3.2 Hybrid Liquidity Architecture

Flow: Internal CLOB Matching → Net Exposure Calculation → Threshold-Based Hedging → Binance/Bybit Execution

- Internal matching takes priority — all orders attempt internal fill first
- Net exposure engine: Net = Total Long – Total Short, calculated per asset in real-time
- Configurable hedge ratio per asset (e.g. hedge 80% of net exposure above threshold)
- Hedging engine connects to Binance (spot + futures) and Bybit (spot + derivatives) via corporate API accounts
- Hedging is position-based — no direct order routing from user to external exchange
- Hedging engine operates as a treasury function — users interact only with the internal CLOB
- Derivatives hedging via Binance/Bybit futures — no direct derivative order routing to end users from external venues

3.3 Fee Engine

The fee engine is a critical revenue and compliance component and requires full specification:

- Maker/taker fee differentiation — default maker 0.10%, taker 0.15% (configurable per instance)
- Tiered VIP fee schedule based on 30-day trailing volume — minimum 5 tiers per instance
- Native token fee discount — users holding/staking the platform token receive a configurable fee reduction
- Referral fee splits — configurable percentage of referred user's trading fees credited to referrer
- Withdrawal fee management — per-asset, per-network flat fee or percentage, updatable by admin without deployment

- INR fiat conversion spread — configurable bid/ask spread on USDT/INR pair with dynamic premium based on USD/INR live rate
- Fee ledger — all collected fees posted to a dedicated fee ledger account, separate from user balances
- Revenue reporting — real-time and historical fee income by category (trading, withdrawal, conversion spread, OTC)
- Fee override per user — admin ability to apply custom fee rates for institutional and VIP accounts

3.4 Settlement & Clearing

Distinct from matching — settlement governs the post-trade lifecycle:

- Real-time P&L mark-to-market for all open positions
- Trade settlement: matched trade → balance debit/credit cycle completed within one processing tick
- End-of-day settlement run: reconcile all matched trades against user balance ledgers
- Cross-asset netting: for users with positions in multiple instruments, net margin exposure across correlated assets
- Margin call processing: automated margin call notification → grace period → partial liquidation → full liquidation pipeline
- Settlement ledger: immutable append-only record of every settlement event, queryable for audit
- Reconciliation engine: daily reconciliation of internal ledger vs Fireblocks custody wallet balances vs exchange hot wallet
- Clearing fund: segregated clearing fund to cover failed settlements in edge cases

4. Trading Modules

4.1 Spot Trading

- Full hybrid model — internal CLOB with Binance/Bybit liquidity backstop
- All order types: Market, Limit, Stop-Limit, Stop-Market, IOC, FOK, Post-Only
- Real-time charts (TradingView charting library integration)
- Order book widget with configurable depth and grouping
- Trade history, open orders, order history panels
- Price alerts — user-configurable alerts via notification system

4.2 Margin Trading

- Cross margin and isolated margin modes
- Borrow/lend system with per-asset interest rate configuration
- Automated interest accrual and deduction from margin balance
- Liquidation engine: mark price-triggered, partial then full liquidation cascade
- Portfolio margining: cross-product margin offsets (e.g. spot BTC long offsets short BTC perp)
- Hybrid liquidity support — margin orders route through internal CLOB first

4.3 Perpetual Futures

- Internal matching with position ledger
- Mark price sourced from external index (weighted average of Binance + Bybit + Coinbase spot)
- Funding rate system: 8-hourly funding, rate calculated from mark vs spot premium
- Liquidation engine with insurance fund backstop
- ADL (Auto De-Leveraging) queue for insurance fund depletion scenarios
- Net exposure hedging — aggregate open interest hedged on Binance/Bybit futures
- Leverage up to 100x — configurable per tier, gated by proficiency questionnaire for retail

4.4 Options

- European-style options (calls and puts) for BTC and ETH initially
- Internal matching engine for options order book
- External routing for strikes/expiries with insufficient internal liquidity
- Black-Scholes mark price model with implied volatility surface
- Greeks display: Delta, Gamma, Vega, Theta per position and portfolio aggregate
- Expiry and settlement logic: cash settlement at expiry based on index price
- Separate margining rules for options positions
- Options chain UI: standard expiry matrix view

4.5 OTC / Institutional Desk

- RFQ (Request for Quote) workflow — institutional user submits size, receives firm two-way quote
- Manual price negotiation interface for relationship manager
- Block trade settlement — OTC trades settled directly to custody without hitting order book
- Dedicated OTC portal for institutional users with enhanced position and P&L reporting
- OTC fee schedule — separate from retail fee tiers, negotiated per relationship
- Minimum block trade size configurable per asset
- OTC trade audit trail — full log of quotes issued, accepted, rejected, and settled

4.6 Trading Bot / Algorithmic Trading

- Full REST and WebSocket API access for algorithmic trading clients
- API rate limits tiered by account level — standard, professional, institutional
- Sub-account structure for bot segregation — each bot/strategy runs in an isolated sub-account
- Order management system (OMS) API: place, amend, cancel, query orders programmatically
- Execution reports via WebSocket: real-time fill notifications with full order state
- FIX protocol support (optional phase 2) for institutional connectivity
- Paper trading / sandbox environment — full API access on test network with simulated liquidity
- API key management: multiple keys per account, IP whitelist per key, read/trade/withdraw permission scoping

5. VIP Account Program

5.1 VIP Tier Structure

Tier	30d Volume (USD)	Maker Fee	Taker Fee	Benefits
Standard	< $50,000	0.10%	0.15%	Standard access
VIP 1	$50k – $500k	0.08%	0.12%	Priority support
VIP 2	$500k – $2M	0.06%	0.10%	+ Higher API limits
VIP 3	$2M – $10M	0.04%	0.08%	+ Dedicated manager
VIP 4	> $10M	Negotiated	Negotiated	+ OTC desk access

5.2 VIP Features

- Automatic tier upgrade/downgrade based on rolling 30-day volume — calculated nightly
- Manual VIP override by admin for institutional onboarding
- Higher withdrawal limits per 24h period
- Dedicated customer support queue with SLA response times
- Preferential API rate limits and WebSocket connection slots
- Early access to new product launches and token listings
- Custom fee negotiation for VIP 4+ accounts
- Sub-account support: VIP users may operate up to 20 sub-accounts under one master account

6. Market Maker Program

- Formal market maker application and onboarding process
- Market maker obligations: minimum quoted depth, maximum spread, minimum uptime percentage (e.g. 95% during market hours)
- Dedicated market maker API endpoints with reduced latency and higher rate limits
- Maker rebate program: negative maker fees (exchange pays market maker) at qualifying volume tiers
- Real-time obligation monitoring dashboard — market maker compliance vs SLA
- Automated rebate calculation and settlement to market maker account daily
- Multiple market maker accounts per firm — segregated by instrument or strategy
- Market maker agreement module: digital agreement acceptance with version tracking
- Liquidity SLA reporting: daily and weekly liquidity quality metrics per instrument

7. INR Fiat System

7.1 INR Wallet

- UPI integration (NPCI-approved payment aggregator) for deposits and withdrawals
- IMPS/NEFT/RTGS bank transfer support
- Internal INR ledger — segregated from crypto ledger
- Daily reconciliation of INR balances vs bank nostro account
- INR withdrawal limits: per-transaction, per-day, per-month — configurable by admin
- TDS deduction engine: automatic 1% TDS deduction on crypto sale proceeds per Indian Income Tax Act Section 194S
- TDS certificate generation for users (Form 16A equivalent)
- GST compliance: 18% GST on trading fees — automated invoice generation

7.2 USDT/INR Conversion Model

- USDT/INR as primary bridge pair — Flow: INR deposit → USDT purchase → crypto trading
- Dynamic pricing: USD/INR live reference rate (RBI/Bloomberg feed) + configurable premium spread
- Internal liquidity: platform acts as principal on USDT/INR conversions — no external routing
- P2P trading module: peer-to-peer INR-USDT trades with escrow and dispute resolution
- Rate lock: user can lock a conversion rate for up to 30 seconds before confirming
- Slippage protection: maximum slippage tolerance configurable per user

8. Custody & Compliance Infrastructure

8.1 Fireblocks (Custody)

- Separate Fireblocks workspace per exchange instance — 4 workspaces total
- Multi-chain wallet support: Bitcoin, Ethereum, BNB Chain, Polygon, Solana, Tron, Arbitrum, Optimism at minimum
- Automated deposit detection and user balance crediting
- Withdrawal queue with Fireblocks policy engine approval workflow
- Hot and cold wallet rebalancing: configurable thresholds — e.g. maintain max 5% of assets in hot wallet
- Automated cold-to-hot transfer when hot wallet falls below minimum threshold
- Multi-signature withdrawal policy for large withdrawals (above configurable threshold)
- Fireblocks webhook integration for real-time transaction status updates

8.2 Chainalysis (KYT)

- Real-time transaction screening on all deposits and withdrawals
- Risk score thresholds: auto-approve (green), manual review (amber), auto-reject (red)
- Sanctions screening: OFAC, UN, EU consolidated lists
- Exposure categories: darknet, mixer, ransomware, high-risk exchange flagging
- Alert queue in admin panel for compliance officer review

8.3 AML & Transaction Monitoring

- Rule-based transaction monitoring engine: velocity rules, structuring detection, unusual pattern alerts
- Risk-based customer scoring: updated dynamically based on transaction behaviour
- Suspicious Transaction Report (STR) workflow: trigger → investigation notes → submission to regulator
- Currency Transaction Report (CTR) for high-value transactions per jurisdiction thresholds
- Peer-to-peer transaction monitoring on P2P module
- FIU-IND integration: automated STR/CTR filing in required format for AGCE India
- VARA/CMA integration: regulatory reporting outputs in required formats for UAE instances

8.4 KYC (Didit)

- Tiered KYC: Level 1 (email + phone) → Level 2 (govt ID + selfie) → Level 3 (enhanced due diligence for VIP/institutional)
- Video KYC (VKYC) for AGCE India — mandatory for INR fiat access
- Jurisdiction-specific document types: Aadhaar/PAN for India, Emirates ID/Passport for UAE
- Re-KYC triggers: periodic refresh (annual for standard, 6-monthly for high-risk)

- KYC status reflected in trading limits and product access

8.5 Travel Rule (CryptoSwift)

- Travel Rule compliance for all crypto transfers above USD 1,000 equivalent
- VASP counterparty identification on outbound transfers
- Originator and beneficiary data transmission to compliant VASPs
- Unhosted wallet declaration workflow for transfers to self-hosted wallets

9. Risk & Hedging Engine

- Real-time exposure tracking per asset, per user, per instrument
- Automated hedging triggered when net exposure exceeds configurable threshold
- Hedge ratio: configurable per asset — e.g. hedge 80% of net exposure above $50,000
- Risk limits: per-user position limits, per-asset concentration limits, platform-wide exposure cap
- Funding rate risk management for perpetuals — alert when funding rate exceeds configurable threshold
- Liquidation cascade protection: maximum liquidation volume per block to prevent market impact
- Insurance fund management: automated replenishment from liquidation surplus
- Stress test scenarios: daily P&L simulation under defined price shock scenarios
- Risk dashboard for treasury team: live exposure, hedge positions, insurance fund level

10. Market Surveillance

- Wash trading detection: identify accounts with self-matched or collusive orders
- Spoofing and layering detection: large orders placed and immediately cancelled at key price levels
- Front-running detection: monitor for order patterns correlated with pending large orders
- Position concentration alerts: flag users approaching platform-wide position caps
- Unusual volume alerts: volume spikes above configurable multiples of 30-day average
- Large trader reporting: automatic flagging of accounts exceeding regulatory reporting thresholds
- Surveillance case management: open/investigate/close workflow with audit trail
- Regulatory reporting: export surveillance findings in regulator-specific formats

11. Notification & Communication System

11.1 Notification Channels

- In-app notification centre — persistent notification history
- Email notifications via transactional email provider (SendGrid or AWS SES)
- SMS notifications via Twilio or equivalent
- Push notifications — iOS and Android mobile app
- Webhook notifications — for API users and trading bots
- Telegram bot integration (optional phase 2)

11.2 Notification Events

- Trade execution: order filled (full and partial), order cancelled, order expired
- Deposit and withdrawal: initiated, confirmed, completed, failed, flagged
- Margin and liquidation: margin call warning (multiple thresholds), liquidation initiated, liquidation completed
- Account security: login from new device, password change, API key created/deleted, withdrawal address added
- Price alerts: user-set price target reached for any instrument
- System events: scheduled maintenance, emergency halt, product launch
- Compliance events: KYC status change, document expiry reminder, account restriction

11.3 Notification Preferences

- User-configurable per event type and per channel
- Do-not-disturb hours configurable
- Language preference — notifications served in user's preferred language

12. API Access

12.1 REST API

- Full trading API: place/amend/cancel orders, query balances, order history, trade history
- Market data API: order book snapshots, ticker, OHLCV, funding rates, mark prices
- Account API: sub-account management, balance transfers, deposit/withdrawal initiation
- OpenAPI 3.0 specification published and version-controlled
- API versioning: /v1/, /v2/ — deprecated versions maintained for minimum 6 months

12.2 WebSocket API

- Real-time order book updates (incremental and full snapshots)
- Trade stream: public trade executions
- User data stream: account balance updates, order execution reports, position updates
- Mark price and funding rate stream for perpetuals
- Ticker stream: 24h stats for all instruments

12.3 API Key Management

- Multiple API keys per account (up to 10 for standard, 50 for institutional)
- Per-key permission scoping: Read, Trade, Withdraw — each independently toggled
- Per-key IP whitelist — key only accepted from specified IP addresses
- Key expiry date — optional automatic expiry
- Key activity log — last used timestamp and IP
- Emergency key revocation — revoke all keys instantly from account security panel

13. Tax Reporting (India-Specific)

- Per-user capital gains report: buy/sell pairs with cost basis (FIFO and AVCO methods)
- TDS deduction ledger: all TDS deducted with dates, amounts, applicable rates
- Annual tax summary export: PDF and CSV formats for user's CA/tax filing
- Form 26AS reconciliation support: TDS amounts reconcilable with income tax portal
- GST invoice generation: per-transaction GST invoice for platform fees
- Admin TDS remittance report: aggregated TDS collected per quarter for remittance to income tax department

14. Additional Product Modules

14.1 Staking & Earn

- Flexible and locked staking products — admin-configurable APY and lock periods
- Staking rewards ledger — accrual and distribution engine
- Earn product marketplace — display available products with yield and risk rating

14.2 P2P Trading

- P2P marketplace for INR-USDT and other fiat pairs
- Escrow engine: crypto held in escrow until fiat payment confirmed
- Dispute resolution workflow: user → automated → manual escalation
- Merchant verification: tiered merchant status with completion rate and volume history
- P2P AML monitoring: suspicious P2P patterns flagged to compliance queue

14.3 Copy Trading

- Strategy provider registration and performance verification
- Follower allocation: copy a % of provider's trades up to follower's configured capital
- Performance fee: configurable split between platform and strategy provider
- Risk controls: follower can set maximum drawdown stop-loss

14.4 Token Launchpad & TGE

- Project application and vetting workflow
- Token sale mechanics: fixed price, Dutch auction, whitelist allocation
- Vesting schedule engine: cliff + linear vesting with on-chain or off-chain tracking
- TGE listing: automatic token listing on spot market post-sale completion

14.5 Referral, Airdrop & Custom Listing

- Referral program: unique referral links, configurable commission rates, tiered bonuses
- Airdrop engine: snapshot-based or campaign-based token distribution
- Custom token listing module: listing application, due diligence checklist, listing fee management

15. Client Portal & Support

- Dedicated user portal: account overview, P&L summary, tax documents, API key management
- Support ticketing system: user submits ticket → automated triage → agent assignment → resolution tracking
- Knowledge base integration: FAQ and help articles linked contextually within the platform
- Live chat integration (e.g. Intercom or Zendesk) for real-time support
- Escalation workflow: L1 → L2 → compliance/legal escalation with SLA timers
- User activity log: users can view their own login history, order history, and device history

16. Admin Panel

The admin panel must provide granular control over all platform functions. Below is the required feature set:

16.1 User Management

- User search by email, UID, wallet address, phone
- KYC status management: approve, reject, escalate
- Account restriction: read-only, trading halt, withdrawal freeze, full suspension
- Fee override: apply custom fee schedule per user
- Balance adjustment: credit/debit user balance with mandatory audit note
- VIP tier manual override

16.2 Risk & Trading Controls

- Global trading halt: pause all trading on one or all instances
- Instrument-level controls: pause, delist, adjust tick size / lot size
- Leverage limit adjustment per instrument and per user tier
- Liquidation engine parameters: maintenance margin, liquidation fee
- Insurance fund view and manual adjustment

16.3 Compliance & Surveillance Dashboard

- AML alert queue with investigation workflow
- KYC review queue
- STR/CTR filing interface
- Surveillance case management
- Regulatory report generation and export

16.4 Finance & Treasury

- Real-time platform P&L: fee income, hedging P&L, conversion spread income
- Treasury position view: hot/cold wallet balances per chain per instance
- Hedging position view: open hedge positions on Binance/Bybit
- Manual withdrawal approval queue for large transactions

16.5 System Configuration

- Fee schedule management: create/update tiered fee structures without deployment
- Asset management: add new chains, tokens, deposit/withdrawal enable/disable per asset
- Announcement and notification broadcast
- Maintenance mode toggle per module

- Admin role management: RBAC with instance-scoped permissions and full audit log

17. Analytics & Business Intelligence

- Real-time trading volume dashboard: by pair, by user segment, by instance
- User acquisition and retention metrics: registration funnel, KYC conversion, first trade conversion
- Liquidity quality analytics: bid-ask spread over time, order book depth history
- Hedging P&L analytics: realized and unrealized P&L on hedge book
- Revenue analytics: fee income breakdown by category and instance
- Risk analytics: exposure concentration, VaR estimates, insurance fund adequacy
- Export: all reports exportable as CSV and PDF with scheduled email delivery

18. Security

- 2FA: TOTP (Google Authenticator) and SMS OTP — 2FA mandatory for withdrawals
- Withdrawal whitelist: addresses must be whitelisted with 24-hour waiting period before first use
- Anti-phishing code: user-set code displayed in all platform emails
- Session management: concurrent session limits, session expiry, force-logout
- Encryption: AES-256 at rest, TLS 1.3 in transit
- DDoS protection: Cloudflare Enterprise or equivalent, rate limiting at CDN and application layer
- Penetration testing: mandatory before go-live, annual thereafter — reports to be provided to regulator on request
- VAPT by qualified third party — Palo Alto Networks Cortex or equivalent for ongoing threat monitoring
- Bug bounty program: responsible disclosure policy published at launch
- Key management: all secrets managed via HashiCorp Vault or cloud-native KMS

19. Infrastructure

- Microservices architecture — each module (matching engine, risk engine, wallet service, notification service, etc.) independently deployable
- Matching engine and risk engine: Go or Rust — low-latency requirement
- Event streaming: Apache Kafka for inter-service messaging
- Caching layer: Redis for order book state, session data, rate limiting
- Primary database: PostgreSQL for transactional data
- Time-series database: InfluxDB or TimescaleDB for OHLCV and metrics data
- Cloud deployment: AWS as primary for India (Mumbai region), Huawei Cloud UAE or AWS UAE for UAE instances
- Container orchestration: Kubernetes with Helm charts for each service
- CI/CD pipeline: automated testing, staging deployment, blue-green production deployments
- Observability: Prometheus + Grafana for metrics, ELK stack for logs, Jaeger for distributed tracing
- SRE: defined SLOs — 99.9% uptime for trading engine, 99.5% for ancillary services
- Disaster recovery: RPO 15 minutes, RTO 1 hour — multi-region failover for each instance
- BCP documentation: written runbooks for all critical failure scenarios

20. Legal & Compliance Framework

- In-app jurisdiction detection at registration — user directed to correct instance based on IP and declared country
- Terms of Service acceptance: electronic signature with timestamp, IP, and device fingerprint logged
- Re-acceptance triggers: ToS version update, material product change, regulatory requirement
- Product eligibility gating: derivatives, margin trading, P2P gated behind jurisdiction and KYC level checks
- Minor detection: date of birth verification — accounts for users below 18 blocked at KYC
- Sanctions re-screening: all users re-screened against updated sanctions lists weekly
- Data retention: 5 years for all transaction records, 10 years for KYC documents — per jurisdiction minimum
- Right to erasure: GDPR-compliant erasure workflow for non-obligated data on AGCE Global

21. Deliverables

#	Deliverable	Scope	Acceptance Criteria
1.	Modern Website/Landing Pages for All 4 instance of exchanges	-	Attractive UI/UX, Responsive, glassmorphism, interactive components, Interactive modern components, not template based but custom, SEO friendly
2.	Web Platform for relevant exchange	-	Above mentioned features & Light/Dark Theme
3.	Android App for relevant Exchange	-	Above mentioned features & Light/Dark Theme
4.	IOS App for relevant exchange	-	Above mentioned features & Light/Dark Theme
1	System architecture documentation	All 4 instances	Reviewed and approved by client
2	Matching engine implementation	Core + all instances	Load test: 50,000 orders/sec, <1ms latency at p99
3	Hedging engine	All instances	Hedge executes within 500ms of threshold breach
4	Jurisdiction guardrails	All 4 instances	Penetration test confirms zero cross-instance data leakage
5	Fee engine	All instances	Fee calculations verified against test matrix
6	Settlement & clearing module	All instances	Reconciliation passes 30-day simulation test
7	INR fiat system (AGCE India)	India instance only	UPI deposit/withdrawal cycle completes end-to-end
8	Custody integration (Fireblocks)	All instances (4 workspaces)	Deposit/withdrawal cycle tested per chain
9	KYC/AML/surveillance stack	All instances	Compliance demo to regulator if required
10	Admin panel	All instances	Full feature walkthrough approved by client

#	Deliverable	Scope	Acceptance Criteria
11	REST & WebSocket APIs	All instances	OpenAPI spec delivered; API test suite passes 100%
12	Load testing report	All instances	50,000 concurrent users simulated per instance
13	Security penetration test report	All instances	Critical and high findings resolved before go-live
14	Source code handover	Full codebase	Escrow deposit + client GitHub access + build documentation
15	Deployment on client infrastructure	All instances	All 4 instances live and accessible
16	Technical documentation	Full platform	Architecture, API, runbook, DR plan delivered

22. Key Constraints

- Must NOT be a Binance or Bybit white-label wrapper — internal matching engine is mandatory
- Hedging must be position-based — no direct order routing of user orders to external exchanges
- INR system must be independently operated — not solely dependent on a third-party provider
- Source code must be fully owned by Arab Global — no vendor lock-in, no obfuscated code
- All four instances must be independently operable — the failure or shutdown of one instance must not affect others
- Regulatory guardrails are non-negotiable — jurisdiction isolation must be enforced at architecture level, not only at policy level

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